

## Communication

|               |                                                    |
|---------------|----------------------------------------------------|
| Date          | 10/07/2026                                         |
| Name          | Karuru Arjun                                       |
| Project Name  | Voice-Based Concept Understanding Analyser (VBCUA) |
| Maximum Marks | 1 Mark                                             |

### Communication Plan:

Effective communication was maintained throughout the development of the Voice-Based Concept Understanding Analyser (VBCUA) project. As an individual project, communication mainly involved project planning, mentor guidance, technical research, documentation updates, and regular progress reviews to ensure timely completion and successful project delivery.

| S.No | Communication Type        | Frequency | Channel / Tool                               | Participants            | Purpose                                                                        |
|------|---------------------------|-----------|----------------------------------------------|-------------------------|--------------------------------------------------------------------------------|
| 1    | Project Planning          | Daily     | Personal Notes, VS Code                      | Self                    | Plan daily development tasks, organize project workflow, and monitor progress. |
| 2    | Progress Review           | Weekly    | GitHub, VS Code                              | Self                    | Review completed modules, test model performance, and identify pending work.   |
| 3    | Issue / Bug Resolution    | As Needed | Stack Overflow, Python Documentation, GitHub | Self                    | Resolve coding errors, dependency issues, and model integration problems.      |
| 4    | Guide / Mentor Discussion | Weekly    | Google Meet / WhatsApp                       | Student & Project Guide | Discuss project progress, clarify doubts, and                                  |

|   |                          |      |                       |      |                                                                           |
|---|--------------------------|------|-----------------------|------|---------------------------------------------------------------------------|
|   |                          |      |                       |      | receive technical guidance.                                               |
| 5 | Documentation Review     | Once | Microsoft Word        | Self | Verify project documentation, screenshots, and reports before submission. |
| 6 | Final Project Submission | Once | Skill Wallet & GitHub | Self | Submit the complete project, source code, report, and required documents. |

### Communication Challenges & Resolutions:

| S.No | Challenge Faced                                                                | Resolution / Action Taken                                                                                      |
|------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1    | Integrating OpenAI Whisper, Sentence-BERT, and Librosa into a single workflow. | Tested each module individually, verified data flow, and integrated all components successfully.               |
| 2    | Managing Python dependencies and package compatibility.                        | Installed required packages using requirements.txt, resolved version conflicts, and verified the environment.. |
| 3    | Generating accurate semantic similarity scores                                 | Used Sentence-BERT embeddings with cosine similarity and validated results using multiple sample inputs.       |